

Video analytics applications

Video analytics systems are widely used in airports, the railways, seaports, industries, retail, etc. Some of the areas where video analytics can be useful are:

Security applications. Security personnel can record only the relevant data and access it quickly when needed. Live viewing also makes it more productive as live security alerts can be configured to notify security personnel when threats are detected on camera.

Transportation. This system allows security professionals to keep an eye on highways, train platforms, toll booths on highways, parking lots, garages and passenger pick-up/drop-off locations.

Healthcare. They help to monitor patients and others present at a hospital, minimising the damage to property. While monitoring restricted hallways, corridors, equipment rooms, emergency rooms, entry points, parking lots, garages and pharmacies, video analytics software can notify the appropriate personnel immediately when threats are located.

Education. Video analytics can help keep schools and educational facilities safe and secure, allowing staff to monitor hallways, entrances, exits, parking lots and entire campuses. It can help keep both students and faculty safe.

Military/policing. These systems enable military personnel and the police to monitor restricted areas. Video analytics can both improve the safety of bystanders as well as law enforcement officials. Military personnel and police can easily monitor locations that are inaccessible or too harmful for humans to enter.

distributed architecture assigns intelligent processing to different nodes of the video surveillance system. This infrastructure helps to reduce wiring costs and the bandwidth required in the video surveillance network.

"Each surveillance camera automatically identifies activities and objects during continuous real-time image analysis, and alerts the operator by sending an alarm when it detects a critical event," informs Ajay Sahney, senior general manager and country head, Bosch Security Systems.

In short, video analytics technology will enable users to manage a vast mass of undifferentiated data that they may receive from different cameras and turn it into useful information. This will turn their cameras into predictive tools that will allow them to spot problems brewing and prevent incidents in a more proactive manner, rather than just filming the events for later investigative use.

Informs Mohit Jindal, senior manager, head-sales & operations, AV Systems, "Most video analytics today run on servers, but accelerated PCs are also getting into the act moving the analytic engines closer to the edge of the network where the video cameras themselves are located."

Benefits of video analytics

Video analytics technology within a CCTV system offers many benefits.

Shares Sanjeev Sehgal, managing director, Samridhi Automations (Spars), "Video analytics uses a variety of rules, which can be specifically customised both for the scene and the objects being observed, in order to intelligently identify potentially suspicious behaviour."

By using multiple customised rules and algorithms that will screen out adverse weather, the effects of changing light conditions, and non-critical movement, analytics systems let security

staff focus on real incidents rather than getting bogged down with hundreds of false alarms. This technology also requires less time to monitor multiple video feeds within your CCTV system and allows quicker responses.

Explains Sahney, "No matter how many cameras a system uses, monitoring everything effectively is a serious challenge. Even by observing one camera for long, an operator can miss out on important activities. Bosch intelligent video analysis (IVA) helps operators to stay focused by introducing a new level of automation to CCTV monitoring. Edge-based, real-time processing identifies alert conditions, giving your security team the information it needs to react swiftly and take action."

This technology reduces the manpower or security personnel required to watch multiple video monitors at the control room. It can be implemented on water as well. Thus any suspicious movement in a lake front or large pool can be easily traced by this system.

"Since video analytics can be used to transmit only specific scenes of interest, the bandwidth required for its use is significantly less. Many of the embedded analytic algorithms can compensate for various camera issues and quality levels, making it possible to use low-cost cameras or achieve better range and resolution. Closely associated with this is the increasing capability of thermal cameras in the outdoor environment leading to improved detection capabilities at night," adds Sehgal.

MAJOR CONTRIBUTORS TO THIS REPORT



Ajay Sahney,
Senior General Manager
and Country Head, Bosch
Security Systems



Mohit Jindal,
Senior Manager, Head-Sales
& Operations,
AV Systems



Ramakrishna Sundaram,
Executive Vice President
and Head, Building
Technologies Division,
Infrastructure and Cities
Sector, Siemens



Sanjeev Sehgal,
Managing Director, Samridhi
Automations (Spars)